

 ARDUINO CODE - Copy this entire code:

```
cpp

/*
 * Digital Quiz Buzzer System
 * Board: Shrike Lite RP2040 with IO Shield
 *
 * Features:
 * - 4 Push Buttons (3 for players, 1 for reset)
 * - Buzzer for audio feedback
 * - LCD display to show winner
 *
 * Components on IO Shield:
 * - Button 1 (Pin 20) → Player 1
 * - Button 2 (Pin 21) → Player 2
 * - Button 3 (Pin 22) → Player 3
 * - Button 4 (Pin 26) → Reset Button
 * - Buzzer (Pin 27)
 * - LCD Display (I2C: SDA-Pin 4, SCL-Pin 5)
 */

// Include the LCD library for I2C communication
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

// Initialize LCD with I2C address 0x27 and 16x2 display size
LiquidCrystal_I2C lcd(0x27, 16, 2);

// Pin definitions for IO Shield
#define BUTTON_1 20 // Player 1 button
#define BUTTON_2 21 // Player 2 button
#define BUTTON_3 22 // Player 3 button
```

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#define RESET_BUTTON 26 // Reset button

#define BUZZER 27 // Buzzer pin


// Game state variables

bool gameActive = true; // Tracks if game is active or someone has won
int winner = 0; // Stores the winner (0 = no winner, 1-3 = player number)


void setup() {
    // Initialize serial communication for debugging
    Serial.begin(9600);


    // Configure button pins as INPUT with internal pull-up resistors
    // Pull-up means button reads HIGH when not pressed, LOW when pressed
    pinMode(BUTTON_1, INPUT_PULLUP);
    pinMode(BUTTON_2, INPUT_PULLUP);
    pinMode(BUTTON_3, INPUT_PULLUP);
    pinMode(RESET_BUTTON, INPUT_PULLUP);


    // Configure buzzer pin as output
    pinMode(BUZZER, OUTPUT);
    digitalWrite(BUZZER, LOW); // Ensure buzzer is off initially


    // Initialize I2C communication for LCD
    Wire.setSDA(4); // Set SDA pin
    Wire.setSCL(5); // Set SCL pin
    Wire.begin();


    // Initialize the LCD
    lcd.init();
    lcd.backlight(); // Turn on LCD backlight
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// Display welcome message
displayWelcome();

Serial.println("Quiz Buzzer System Ready!");
}

void loop() {
    // Check reset button (active when someone has won)
    if (digitalRead(RESET_BUTTON) == LOW) {
        delay(50); // Debounce delay
        if (digitalRead(RESET_BUTTON) == LOW) {
            resetGame();
            while(digitalRead(RESET_BUTTON) == LOW); // Wait for button release
        }
    }

    // Only check player buttons if game is still active
    if (gameActive) {

        // Check Player 1 button
        if (digitalRead(BUTTON_1) == LOW) {
            delay(50); // Debounce delay
            if (digitalRead(BUTTON_1) == LOW && gameActive) {
                playerWins(1);
            }
        }

        // Check Player 2 button
        if (digitalRead(BUTTON_2) == LOW) {
            delay(50); // Debounce delay
            if (digitalRead(BUTTON_2) == LOW && gameActive) {

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        playerWins(2);
    }
}

// Check Player 3 button
if (digitalRead(BUTTON_3) == LOW) {
    delay(50); // Debounce delay
    if (digitalRead(BUTTON_3) == LOW && gameActive) {
        playerWins(3);
    }
}
}

// Function called when a player wins
void playerWins(int playerNumber) {
    gameActive = false; // Disable further button presses
    winner = playerNumber;

    // Sound the buzzer
    beepBuzzer();

    // Display winner on LCD
    displayWinner(playerNumber);

    // Print to serial monitor for debugging
    Serial.print("Player ");
    Serial.print(playerNumber);
    Serial.println(" wins!");
}

```

```
// Function to beep the buzzer
```

```
void beepBuzzer() {  
    digitalWrite(BUZZER, HIGH);  
    delay(200); // Beep for 200ms  
    digitalWrite(BUZZER, LOW);  
    delay(100);  
    digitalWrite(BUZZER, HIGH);  
    delay(200); // Second beep  
    digitalWrite(BUZZER, LOW);  
}
```

```
// Function to display winner on LCD
```

```
void displayWinner(int playerNumber) {  
    lcd.clear();  
    lcd.setCursor(0, 0);  
    lcd.print(" WINNER!");  
    lcd.setCursor(0, 1);  
    lcd.print(" Player ");  
    lcd.print(playerNumber);  
}
```

```
// Function to display welcome message
```

```
void displayWelcome() {  
    lcd.clear();  
    lcd.setCursor(0, 0);  
    lcd.print(" Quiz Buzzer");  
    lcd.setCursor(0, 1);  
    lcd.print("Press to Start!");  
}
```

```
// Function to reset the game
```

```

void resetGame() {
    gameActive = true; // Re-enable game
    winner = 0;      // Clear winner

    // Short beep to confirm reset
    digitalWrite(BUZZER, HIGH);
    delay(100);
    digitalWrite(BUZZER, LOW);

    // Display welcome message again
    displayWelcome();

    Serial.println("Game Reset!");
}
` ``

```

🍷 CONNECTION (Very Simple!)

****NO WIRING NEEDED!**** Just stack the IO Shield on top of the Shrike Lite RP2040 board. All buttons, buzzer, and LCD are already connected.

📺 SETUP STEPS:

1. Install Library in Arduino IDE:

- Open Arduino IDE
- Go to: `Sketch` → `Include Library` → `Manage Libraries`
- Search for: *****LiquidCrystal I2C*****

- Install the one by ****Frank de Brabander****

2. Install RP2040 Board:

- Go to `File` → `Preferences`

- In "Additional Board Manager URLs", paste:

```\n

[https://github.com/earlephilhower/arduino-pico/releases/download/global/package\\_rp2040\\_index.json](https://github.com/earlephilhower/arduino-pico/releases/download/global/package_rp2040_index.json)

- Go to Tools → Board → Boards Manager
- Search for "**rp2040**"
- Install "**Raspberry Pi Pico/RP2040**"

**3. Select Board:**

- Tools → Board → Raspberry Pi RP2040 Boards → Raspberry Pi Pico

**4. Upload Code:**

- Copy the code above
- Paste into Arduino IDE
- Save as quiz\_buzzer\_system.ino
- Connect USB cable
- Click Upload button (→)

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## HOW TO USE:

1. **Button 1** = Player 1
2. **Button 2** = Player 2
3. **Button 3** = Player 3
4. **Button 4** = Reset

When a player presses first → Buzzer beeps + LCD shows "WINNER! Player X"